EXPERIMENT 8

Configuring the Loan Application with Role-Based Access Control (RBAC) in Zoho Creator

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Subject: Low Code Application Development using Zoho Creator
Project: Bank Loan Application & Pre-Screening Management

Title

Design and implementation of user permissions, roles, role hierarchy and data sharing for the Loan Application & Pre-Screening Management application.

Aim

To give each group of users only the access its work needs, by defining permission profiles, a three-level role hierarchy and a data-sharing rule in Zoho Creator.

Introduction

Loan records contain identity, income and document data. Not every user should see or change all of it. Zoho Creator controls access in layers: **permissions** say what a user may do with each component (view, add, edit, delete, export); **roles** place users in a hierarchy; and **data sharing** decides which records a role can reach. Together these implement the principle of least privilege.

Roles and Permission Profiles

Table 8.1: Roles, hierarchy and permission profiles (assumed design for this experiment)

Role	Position in hierarchy	Permission profile	What the user can do
Branch Manager	Top	Manager	View all records and reports; approve; export
Credit Analyst	Second level (reports to Branch Manager)	Analyst	View applications; edit credit-check fields; no delete
Loan Officer	Third level (reports to Credit Analyst)	Officer	Add and edit applications; update document status; no export, no delete

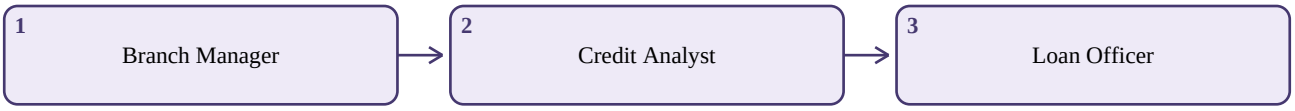


Fig 8.1 — Role hierarchy (top to bottom) (schematic drawn from the specification; not a screenshot)

Procedure

1. Open the Users and permissions area of the application (the labels may differ slightly between versions of Zoho Creator).
2. Create the three permission profiles of Table 8.1 and set, for each component (Loan_Application_Form, Loan_Application_Report, Loan_Command_Center_Page), the allowed actions.
3. Create the roles Branch Manager, Credit Analyst and Loan Officer and arrange them as a hierarchy.
4. Add test users (or invite additional e-mail addresses), assign each a role and a profile.
5. Open the data-sharing settings and create a rule that gives the Credit Analyst role **read-only** access to the records created by Loan Officers.

6 Log in as each test user (or use the preview-as option if available) and check what is visible and editable.

7 Record the results in the verification table.

Permission Matrix

Table 8.2: Permission matrix (design)

Component	Branch Manager	Credit Analyst	Loan Officer
Loan_Application_Form	View, Add, Edit	View, Add	View, Add, Edit
Loan_Application_Report	View, Edit, Export	View, Edit (credit fields)	View, Edit (document fields)
Loan_Command_Center_Page	Open	Open	Open
Branches (master form)	View, Add, Edit	View	View
Delete records	Allowed	Not allowed	Not allowed

Data Sharing Rule

Table 8.3: Data sharing rule

Parameter	Value
Rule name	Analyst_ReadOnly_On_Officer_Records
Form	Loan_Application_Form
Records shared	Records created by users with the Loan Officer role
Shared with	Credit Analyst role
Access level	Read only

Verification (to be completed during the experiment)

Table 8.4: Access verification record

Check	Expected	Observed
Loan Officer opens the report	Can add and edit; Export button absent	
Loan Officer tries to delete a record	Option unavailable	
Credit Analyst opens an officer’s record	Visible, read-only	
Credit Analyst edits a credit-check field	Allowed	
Branch Manager exports the report	Allowed	

Output

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Permission profiles list with the three profiles

Fig 8.2 — Permission profiles

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Role hierarchy showing Branch Manager, Credit Analyst and Loan Officer

Fig 8.3 — Role hierarchy

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Data sharing rule configuration (read-only)

Fig 8.4 — Data sharing rule

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Application as seen by a Loan Officer test user compared with a Branch Manager

Fig 8.5 — Access verified for two roles

Result

Three roles, three permission profiles and a read-only data-sharing rule were configured so that each group of users reaches only the loan data its work requires.

Conclusion

Layered access control keeps the design simple: permissions decide actions, roles give structure, and data sharing opens records across roles only where needed. Checking the application as each role is the proof that the configuration works.